例题：薄板弯曲有限元分析

正方形薄板（如图 1 所示）边长 2*l*=2m，厚度 *t*=0.01m，材料弹性模量 *E*=200GPa，泊松比 μ=0.3。薄板四边固支，其中心受垂直集中载荷 *P*=400N 作用，试求薄板中心的挠度和中心下表面的应力。

【最大挠度理论解 $w=-0.0056\dfrac{P(2l)^2}{\bar{D}}=-0.00048922\ \mathrm{m}$，　$\bar{D}=\dfrac{Et^3}{12(1-\mu^2)}$ 】

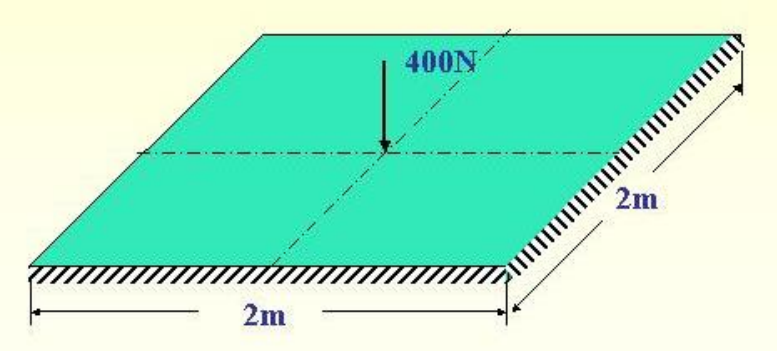
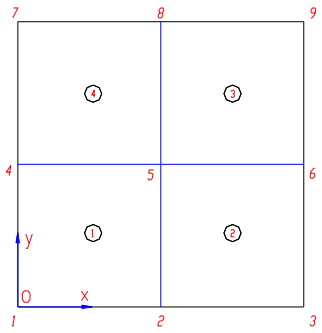


图 1　薄板结构

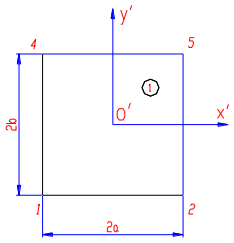
解：求解中采用国际单位制。

单元划分如下图所示，整体结点位移向量：

$$\{\delta\}=\{w_1,\theta_{x1},\theta_{y1},w_2,\theta_{x2},\theta_{y2},w_3,\theta_{x3},\theta_{y3},w_4,\theta_{x4},\theta_{y4},w_5,\theta_{x5},\theta_{y5},w_6,\theta_{x6},\theta_{y6},w_7,\theta_{x7},\theta_{y7},w_8,\theta_{x8},\theta_{y8},w_9,\theta_{x9},\theta_{y9},\}^T$$



整体坐标系



单元局部坐标系

由于此问题的单元坐标系与整体坐标系为平移关系，所以在两种坐标系中单元的刚度方程不变。

在单元坐标系中，薄板单元的刚度矩阵计算公式为：

$$[K]^e=\frac{t^3}{12}\iint_A[B]^T[D][B]dxdy$$

式中，$[B]=\begin{bmatrix}B_i & B_j & B_m & B_p\end{bmatrix}$，

$$[B_k]=-\begin{bmatrix}\frac{1}{8}\left(-\frac{6x}{x_i^3}\right)\left(1+\frac{y}{y_i}\right) & 0 & \frac{x_i}{8}\left(1+\frac{y}{y_i}\right)\left(-\frac{2}{x_i^2}-\frac{6x}{x_i^3}\right)\\ \frac{1}{8}\left(-\frac{6y}{y_i^3}\right)\left(1+\frac{x}{x_i}\right) & -\frac{y_i}{8}\left(1+\frac{x}{x_i}\right)\left(-\frac{2}{y_i^2}-\frac{6y}{y_i^3}\right) & 0\\ \frac{1}{4}\left(\frac{2}{x_iy_i}+\frac{1}{y_i}\left(\frac{1}{x_i}-\frac{3x^2}{x_i^3}\right)+\frac{1}{x_i}\left(\frac{1}{y_i}-\frac{3y^2}{y_i^3}\right)\right) & -\frac{y_i}{4x_i}\left(\frac{1}{y_i}-\frac{2y}{y_i^2}-\frac{3y^2}{y_i^3}\right) & \frac{x_i}{4y_i}\left(\frac{1}{x_i}-\frac{2x}{x_i^2}-\frac{3x^2}{x_i^3}\right)\end{bmatrix}(k=i,j,m,p)$$

$$[D]=\frac{E}{1-\mu^2}\begin{bmatrix}1 & \mu & 0\\ \mu & 1 & 0\\ 0 & 0 & \frac{1-\mu}{2}\end{bmatrix}$$

由上式可计算各单元的刚度矩阵：

单元 1 刚度矩阵：

$$[K]_1^e=10^5\begin{bmatrix}1.9341 & 0.4469 & -0.4469 & -0.8352 & 0.1026 & -0.3919 & -0.2637 & 0.1575 & -0.1575 & -0.8352 & 0.3919 & -0.1026\\ 0.4469 & 0.2784 & -0.0549 & 0.1026 & 0.0879 & 0 & -0.1575 & 0.0696 & 0 & -0.3919 & 0.1136 & 0\\ -0.4469 & -0.0549 & 0.2784 & 0.3919 & 0 & 0.1136 & 0.1575 & 0 & 0.0696 & -0.1026 & 0 & 0.0879\\ -0.8352 & 0.1026 & 0.3919 & 1.9341 & 0.4469 & 0.4469 & -0.8352 & 0.3919 & 0.1026 & -0.2637 & 0.1575 & 0.1575\\ 0.1026 & 0.0879 & 0 & 0.4469 & 0.2784 & 0.0549 & -0.3919 & 0.1136 & 0 & -0.1575 & 0.0696 & 0\\ -0.3919 & 0 & 0.1136 & 0.4469 & 0.0549 & 0.2784 & 0.1026 & 0 & 0.0879 & -0.1575 & 0 & 0.0696\\ -0.2637 & -0.1575 & 0.1575 & -0.8352 & -0.3919 & 0.1026 & 1.9341 & -0.4469 & 0.4469 & -0.8352 & -0.1026 & 0.3919\\ 0.1575 & 0.0696 & 0 & 0.3919 & 0.1136 & 0 & -0.4469 & 0.2784 & -0.0549 & -0.1026 & 0.0879 & 0\\ -0.1575 & 0 & 0.0696 & 0.1026 & 0 & 0.0879 & 0.4469 & -0.0549 & 0.2784 & -0.3919 & 0 & 0.1136\\ -0.8352 & -0.3919 & -0.1026 & -0.2637 & -0.1575 & -0.1575 & -0.8352 & -0.1026 & -0.3919 & 1.9341 & -0.4469 & -0.4469\\ 0.3919 & 0.1136 & 0 & 0.1575 & 0.0696 & 0 & -0.1026 & 0.0879 & 0 & -0.4469 & 0.2784 & 0.0549\\ -0.1026 & 0 & 0.0879 & 0.1575 & 0 & 0.0696 & 0.3919 & 0 & 0.1136 & -0.4469 & 0.0549 & 0.2784\end{bmatrix}$$

在整体刚度矩阵的贡献自由度为 $(w_1,\theta_{x1},\theta_{y1},w_2,\theta_{x2},\theta_{y2},w_5,\theta_{x5},\theta_{y5},w_4,\theta_{x4},\theta_{y4})$

单元 2 刚度矩阵：

$$[K]_2^e = 10^5 \begin{bmatrix}
1.9341 & 0.4469 & -0.4469 & -0.8352 & 0.1026 & -0.3919 & -0.2637 & 0.1575 & -0.1575 & -0.8352 & 0.3919 & -0.1026 \\
0.4469 & 0.2784 & -0.0549 & 0.1026 & 0.0879 & 0 & -0.1575 & 0.0696 & 0 & -0.3919 & 0.1136 & 0 \\
-0.4469 & -0.0549 & 0.2784 & 0.3919 & 0 & 0.1136 & 0.1575 & 0 & 0.0696 & -0.1026 & 0 & 0.0879 \\
-0.8352 & 0.1026 & 0.3919 & 1.9341 & 0.4469 & 0.4469 & -0.8352 & 0.3919 & 0.1026 & -0.2637 & 0.1575 & 0.1575 \\
0.1026 & 0.0879 & 0 & 0.4469 & 0.2784 & 0.0549 & -0.3919 & 0.1136 & 0 & -0.1575 & 0.0696 & 0 \\
-0.3919 & 0 & 0.1136 & 0.4469 & 0.0549 & 0.2784 & 0.1026 & 0 & 0.0879 & -0.1575 & 0 & 0.0696 \\
-0.2637 & -0.1575 & 0.1575 & -0.8352 & -0.3919 & 0.1026 & 1.9341 & -0.4469 & 0.4469 & -0.8352 & -0.1026 & 0.3919 \\
0.1575 & 0.0696 & 0 & 0.3919 & 0.1136 & 0 & -0.4469 & 0.2784 & -0.0549 & -0.1026 & 0.0879 & 0 \\
-0.1575 & 0 & 0.0696 & 0.1026 & 0 & 0.0879 & 0.4469 & -0.0549 & 0.2784 & -0.3919 & 0 & 0.1136 \\
-0.8352 & -0.3919 & -0.1026 & -0.2637 & -0.1575 & -0.1575 & -0.8352 & -0.1026 & -0.3919 & 1.9341 & -0.4469 & -0.4469 \\
0.3919 & 0.1136 & 0 & 0.1575 & 0.0696 & 0 & -0.1026 & 0.0879 & 0 & -0.4469 & 0.2784 & 0.0549 \\
-0.1026 & 0 & 0.0879 & 0.1575 & 0 & 0.0696 & 0.3919 & 0 & 0.1136 & -0.4469 & 0.0549 & 0.2784
\end{bmatrix}$$

在整体刚度矩阵的贡献自由度为 $(w_2, \theta_{x2}, \theta_{y2}, w_3, \theta_{x3}, \theta_{y3}, w_6, \theta_{x6}, \theta_{y6}, w_5, \theta_{x5}, \theta_{y5})$

单元 3 刚度矩阵：

$$[K]_3^e = 10^5 \begin{bmatrix}
1.9341 & 0.4469 & -0.4469 & -0.8352 & 0.1026 & -0.3919 & -0.2637 & 0.1575 & -0.1575 & -0.8352 & 0.3919 & -0.1026 \\
0.4469 & 0.2784 & -0.0549 & 0.1026 & 0.0879 & 0 & -0.1575 & 0.0696 & 0 & -0.3919 & 0.1136 & 0 \\
-0.4469 & -0.0549 & 0.2784 & 0.3919 & 0 & 0.1136 & 0.1575 & 0 & 0.0696 & -0.1026 & 0 & 0.0879 \\
-0.8352 & 0.1026 & 0.3919 & 1.9341 & 0.4469 & 0.4469 & -0.8352 & 0.3919 & 0.1026 & -0.2637 & 0.1575 & 0.1575 \\
0.1026 & 0.0879 & 0 & 0.4469 & 0.2784 & 0.0549 & -0.3919 & 0.1136 & 0 & -0.1575 & 0.0696 & 0 \\
-0.3919 & 0 & 0.1136 & 0.4469 & 0.0549 & 0.2784 & 0.1026 & 0 & 0.0879 & -0.1575 & 0 & 0.0696 \\
-0.2637 & -0.1575 & 0.1575 & -0.8352 & -0.3919 & 0.1026 & 1.9341 & -0.4469 & 0.4469 & -0.8352 & -0.1026 & 0.3919 \\
0.1575 & 0.0696 & 0 & 0.3919 & 0.1136 & 0 & -0.4469 & 0.2784 & -0.0549 & -0.1026 & 0.0879 & 0 \\
-0.1575 & 0 & 0.0696 & 0.1026 & 0 & 0.0879 & 0.4469 & -0.0549 & 0.2784 & -0.3919 & 0 & 0.1136 \\
-0.8352 & -0.3919 & -0.1026 & -0.2637 & -0.1575 & -0.1575 & -0.8352 & -0.1026 & -0.3919 & 1.9341 & -0.4469 & -0.4469 \\
0.3919 & 0.1136 & 0 & 0.1575 & 0.0696 & 0 & -0.1026 & 0.0879 & 0 & -0.4469 & 0.2784 & 0.0549 \\
-0.1026 & 0 & 0.0879 & 0.1575 & 0 & 0.0696 & 0.3919 & 0 & 0.1136 & -0.4469 & 0.0549 & 0.2784
\end{bmatrix}$$

在整体刚度矩阵的贡献自由度为 $(w_5, \theta_{x5}, \theta_{y5}, w_6, \theta_{x6}, \theta_{y6}, w_9, \theta_{x9}, \theta_{y9}, w_8, \theta_{x8}, \theta_{y8})$

单元 4 刚度矩阵：

$$[K]_4^e = 10^5 \begin{bmatrix}
1.9341 & 0.4469 & -0.4469 & -0.8352 & 0.1026 & -0.3919 & -0.2637 & 0.1575 & -0.1575 & -0.8352 & 0.3919 & -0.1026 \\
0.4469 & 0.2784 & -0.0549 & 0.1026 & 0.0879 & 0 & -0.1575 & 0.0696 & 0 & -0.3919 & 0.1136 & 0 \\
-0.4469 & -0.0549 & 0.2784 & 0.3919 & 0 & 0.1136 & 0.1575 & 0 & 0.0696 & -0.1026 & 0 & 0.0879 \\
-0.8352 & 0.1026 & 0.3919 & 1.9341 & 0.4469 & 0.4469 & -0.8352 & 0.3919 & 0.1026 & -0.2637 & 0.1575 & 0.1575 \\
0.1026 & 0.0879 & 0 & 0.4469 & 0.2784 & 0.0549 & -0.3919 & 0.1136 & 0 & -0.1575 & 0.0696 & 0 \\
-0.3919 & 0 & 0.1136 & 0.4469 & 0.0549 & 0.2784 & 0.1026 & 0 & 0.0879 & -0.1575 & 0 & 0.0696 \\
-0.2637 & -0.1575 & 0.1575 & -0.8352 & -0.3919 & 0.1026 & 1.9341 & -0.4469 & 0.4469 & -0.8352 & -0.1026 & 0.3919 \\
0.1575 & 0.0696 & 0 & 0.3919 & 0.1136 & 0 & -0.4469 & 0.2784 & -0.0549 & -0.1026 & 0.0879 & 0 \\
-0.1575 & 0 & 0.0696 & 0.1026 & 0 & 0.0879 & 0.4469 & -0.0549 & 0.2784 & -0.3919 & 0 & 0.1136 \\
-0.8352 & -0.3919 & -0.1026 & -0.2637 & -0.1575 & -0.1575 & -0.8352 & -0.1026 & -0.3919 & 1.9341 & -0.4469 & -0.4469 \\
0.3919 & 0.1136 & 0 & 0.1575 & 0.0696 & 0 & -0.1026 & 0.0879 & 0 & -0.4469 & 0.2784 & 0.0549 \\
-0.1026 & 0 & 0.0879 & 0.1575 & 0 & 0.0696 & 0.3919 & 0 & 0.1136 & -0.4469 & 0.0549 & 0.2784
\end{bmatrix}$$

在整体刚度矩阵的贡献自由度为 $(w_4, \theta_{x4}, \theta_{y4}, w_5, \theta_{x5}, \theta_{y5}, w_8, \theta_{x8}, \theta_{y8}, w_7, \theta_{x7}, \theta_{y7})$

将单元刚度矩阵整合为整体刚度矩阵为

$$[K]=10^5\begin{bmatrix}
1.9341 & 0.4469 & -0.4469 & -0.8352 & 0.1026 & -0.3919 & 0 & 0 & 0 & -0.8352 & 0.3919 & -0.1026 & -0.2637 & 0.1575 & -0.1575 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0.4469 & 0.2784 & -0.0549 & 0.1026 & 0.0879 & 0 & 0 & 0 & 0 & -0.3919 & 0.1136 & 0 & -0.1575 & 0.0696 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-0.4469 & -0.0549 & 0.2784 & 0.3919 & 0 & 0.1136 & 0 & 0 & 0 & -0.1026 & 0 & 0.0879 & 0.1575 & 0 & 0.0696 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-0.8352 & 0.1026 & 0.3919 & 3.8681 & 0.8938 & 0 & -0.8352 & 0.1026 & -0.3919 & -0.2637 & 0.1575 & 0.1575 & -1.6703 & 0.7839 & 0 & -0.2637 & 0.1575 & -0.1575 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0.1026 & 0.0879 & 0 & 0.8938 & 0.5568 & 0 & 0.1026 & 0.0879 & 0 & -0.1575 & 0.0696 & 0 & -0.7839 & 0.2271 & 0 & -0.1575 & 0.0696 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-0.3919 & 0 & 0.1136 & 0 & 0 & 0.5568 & 0.3919 & 0 & 0.1136 & -0.1575 & 0 & 0.0696 & 0 & 0 & 0.1758 & 0.1575 & 0 & 0.0696 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -0.8352 & 0.1026 & 0.3919 & 1.9341 & 0.4469 & 0.4469 & 0 & 0 & 0 & -0.2637 & 0.1575 & 0.1575 & -0.8352 & 0.3919 & 0.1026 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0.1026 & 0.0879 & 0 & 0.4469 & 0.2784 & 0.0549 & 0 & 0 & 0 & -0.1575 & 0.0696 & 0 & -0.3919 & 0.1136 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -0.3919 & 0 & 0.1136 & 0.4469 & 0.0549 & 0.2784 & 0 & 0 & 0 & -0.1575 & 0 & 0.0696 & 0.1026 & 0 & 0.0879 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-0.8352 & -0.3919 & -0.1026 & -0.2637 & -0.1575 & -0.1575 & 0 & 0 & 0 & 3.8681 & 0 & -0.8938 & -1.6703 & 0 & -0.7839 & 0 & 0 & 0 & -0.8352 & 0.3919 & -0.1026 & -0.2637 & 0.1575 & -0.1575 & 0 & 0 & 0 \\
0.3919 & 0.1136 & 0 & 0.1575 & 0.0696 & 0 & 0 & 0 & 0 & 0 & 0.5568 & 0 & 0 & 0.1758 & 0 & 0 & 0 & 0 & -0.3919 & 0.1136 & 0 & -0.1575 & 0.0696 & 0 & 0 & 0 & 0 \\
-0.1026 & 0 & 0.0879 & 0.1575 & 0 & 0.0696 & 0 & 0 & 0 & -0.8938 & 0 & 0.5568 & 0.7839 & 0 & 0.2271 & 0 & 0 & 0 & -0.1026 & 0 & 0.0879 & 0.1575 & 0 & 0.0696 & 0 & 0 & 0 \\
-0.2637 & -0.1575 & 0.1575 & -1.6703 & -0.7839 & 0 & -0.2637 & -0.1575 & -0.1575 & -1.6703 & 0 & 0.7839 & 7.7363 & 0 & 0 & -1.6703 & 0 & -0.7839 & -0.2637 & 0.1575 & 0.1575 & -1.6703 & 0.7839 & 0 & -0.2637 & 0.1575 & -0.1575 \\
0.1575 & 0.0696 & 0 & 0.7839 & 0.2271 & 0 & 0.1575 & 0.0696 & 0 & 0 & 0.1758 & 0 & 0 & 1.1136 & 0 & 0 & 0.1758 & 0 & -0.1575 & 0.0696 & 0 & -0.7839 & 0.2271 & 0 & -0.1575 & 0.0696 & 0 \\
-0.1575 & 0 & 0.0696 & 0 & 0 & 0.1758 & 0.1575 & 0 & 0.0696 & -0.7839 & 0 & 0.2271 & 0 & 0 & 1.1136 & 0.7839 & 0 & 0.2271 & -0.1575 & 0 & 0.0696 & 0 & 0 & 0.1758 & 0.1575 & 0 & 0.0696 \\
0 & 0 & 0 & -0.2637 & -0.1575 & 0.1575 & -0.8352 & -0.3919 & 0.1026 & 0 & 0 & 0 & -1.6703 & 0 & 0.7839 & 3.8681 & 0 & 0.8938 & 0 & 0 & 0 & -0.2637 & 0.1575 & 0.1575 & -0.8352 & 0.3919 & 0.1026 \\
0 & 0 & 0 & 0.1575 & 0.0696 & 0 & 0.3919 & 0.1136 & 0 & 0 & 0 & 0 & 0 & 0.1758 & 0 & 0 & 0.5568 & 0 & 0 & 0 & 0 & -0.1575 & 0.0696 & 0 & -0.3919 & 0.1136 & 0 \\
0 & 0 & 0 & -0.1575 & 0 & 0.0696 & 0.1026 & 0 & 0.0879 & 0 & 0 & 0 & -0.7839 & 0 & 0.2271 & 0.8938 & 0 & 0.5568 & 0 & 0 & 0 & -0.1575 & 0 & 0.0696 & 0.1026 & 0 & 0.0879 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.8352 & -0.3919 & -0.1026 & -0.2637 & -0.1575 & -0.1575 & 0 & 0 & 0 & 1.9341 & -0.4469 & -0.4469 & -0.8352 & -0.1026 & -0.3919 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.3919 & 0.1136 & 0 & 0.1575 & 0.0696 & 0 & 0 & 0 & 0 & -0.4469 & 0.2784 & 0.0549 & -0.1026 & 0.0879 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.1026 & 0 & 0.0879 & 0.1575 & 0 & 0.0696 & 0 & 0 & 0 & -0.4469 & 0.0549 & 0.2784 & 0.3919 & 0 & 0.1136 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.2637 & -0.1575 & 0.1575 & -1.6703 & -0.7839 & 0 & -0.2637 & -0.1575 & -0.1575 & -0.8352 & -0.1026 & 0.3919 & 3.8681 & -0.8938 & 0 & -0.8352 & -0.1026 & -0.3919 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.1575 & 0.0696 & 0 & 0.7839 & 0.2271 & 0 & 0.1575 & 0.0696 & 0 & -0.1026 & 0.0879 & 0 & -0.8938 & 0.5568 & 0 & -0.1026 & 0.0879 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.1575 & 0 & 0.0696 & 0 & 0 & 0.1758 & 0.1575 & 0 & 0.0696 & -0.3919 & 0 & 0.1136 & 0 & 0 & 0.5568 & 0.3919 & 0 & 0.1136 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.2637 & -0.1575 & 0.1575 & -0.8352 & -0.3919 & 0.1026 & 0 & 0 & 0 & -0.8352 & -0.1026 & 0.3919 & 1.9341 & -0.4469 & 0.4469 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.1575 & 0.0696 & 0 & 0.3919 & 0.1136 & 0 & 0 & 0 & 0 & -0.1026 & 0.0879 & 0 & -0.4469 & 0.2784 & -0.0549 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.1575 & 0 & 0.0696 & 0.1026 & 0 & 0.0879 & 0 & 0 & 0 & -0.3919 & 0 & 0.1136 & 0.4469 & -0.0549 & 0.2784
\end{bmatrix}$$

此问题的位移边界条件：

结点 1，2，3，4，6，7，8，9 固支，相应自由度位移均为 0

结点载荷为：

$$f_{z5}=-400$$

故整体刚度方程为：

$$[K]\begin{Bmatrix}0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\w_5\\\theta_{x5}\\\theta_{y5}\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\end{Bmatrix}=\begin{Bmatrix}f_{z1}\\M_{x1}\\M_{y1}\\f_{z2}\\M_{x2}\\M_{y2}\\f_{z3}\\M_{x3}\\M_{y3}\\f_{z4}\\M_{x4}\\M_{y4}\\-400\\0\\0\\f_{z6}\\M_{x6}\\M_{y6}\\f_{z7}\\M_{x7}\\M_{y7}\\f_{z8}\\M_{x8}\\M_{y8}\\f_{z9}\\M_{x9}\\M_{y9}\end{Bmatrix}$$

划去 0 位移所对应的行和列得到降阶方程为：

$$10^5\begin{bmatrix}7.7363 & 0 & 0\\ 0 & 1.1136 & 0\\ 0 & 0 & 1.1136\end{bmatrix}\begin{Bmatrix}w_5\\ \theta_{x5}\\ \theta_{y5}\end{Bmatrix}=\begin{Bmatrix}-400\\ 0\\ 0\end{Bmatrix}$$

解得：$w_5=-5.1705\times10^{-4},\ \theta_{x5}=0,\ \theta_{y5}=0$

故$\{\delta\}=\{0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ -5.1705\times10^{-4}\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\}^T$

即薄板中心挠度为$-5.1705\times10^{-4}m$

用单元 1 计算薄板中心下表面的应力的结点位移：

单元应力计算公式：$\{\sigma\}=z[D][B]\{\delta\}^e$

将单元 1 的结点位移$\{\delta\}^e=\{0\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\ \ -5.1705\times10^{-4}\ \ 0\ \ 0\ \ 0\ \ 0\ \ 0\}^T$和

z=-0.005 代入单元应力计算公式中，得：

$$\{\sigma\}=\begin{Bmatrix}\sigma_x\\ \sigma_y\\ \tau_{xy}\end{Bmatrix}=\begin{Bmatrix}4.432\times10^6\\ 4.432\times10^6\\ 3.977\times10^5\end{Bmatrix}$$

作业：分别用单元 2，单元 3，单元 4 计算薄板中心下表面的应力，并分析比较。